

A Hybrid AI-Driven Recommendation System for Insurance Product Optimization: A Multi-Layer Approach to Enhancing Agent Productivity and Client Engagement

Abstract

This paper presents a novel hybrid recommendation system designed for the insurance industry, combining rule-based logic, machine learning algorithms, and large language model validation to optimize product recommendations and automate client communication. The proposed three-layer architecture addresses critical challenges in insurance agent productivity while ensuring regulatory compliance and contextual appropriateness. Our system demonstrates significant improvements in recommendation accuracy and generates personalized marketing content, as validated through real-world deployment during an industry hackathon. The research contributes to the growing field of AI applications in financial services, particularly in addressing the complex requirements of insurance product recommendation systems.

Keywords: Hybrid recommendation systems, Insurance technology, Machine learning, Natural language processing, Multi-layer architecture

1. Introduction

The insurance industry faces mounting pressure to digitize operations while maintaining regulatory compliance and personalized customer service. Traditional approaches to product recommendation rely heavily on agent expertise and manual processes, resulting in inconsistent service quality and limited scalability (Johnson et al., 2023). Recent advances in artificial intelligence present opportunities to enhance these processes while preserving the human element crucial to insurance relationships.

1.1 Problem Statement

Insurance agents encounter several challenges in their daily operations:

- Time-intensive manual research for appropriate product recommendations
- Inconsistent communication quality across client interactions
- Difficulty in identifying cross-selling opportunities
- Regulatory compliance requirements in recommendation processes
- Scalability limitations in personalized service delivery

1.2 Research Objectives

This research aims to:

- 1. Develop a hybrid recommendation system that combines multiple AI approaches for superior accuracy**
- 2. Implement automated pitch generation maintaining professional communication standards**
- 3. Create a scalable solution that enhances rather than replaces agent expertise**
- 4. Demonstrate measurable improvements in operational efficiency**

1.3 Contributions

Our primary contributions include:

- A novel three-layer hybrid architecture for insurance product recommendations**
 - Integration of large language models for contextual validation and pitch generation**
 - Empirical evaluation demonstrating superior performance over single-method approaches**
 - A production-ready implementation validated in real-world conditions**
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2. Literature Review and Related Work

2.1 Recommendation Systems in Financial Services

Recommendation systems have found widespread application in e-commerce and entertainment, but their implementation in financial services presents unique challenges (Chen et al., 2022). Insurance product recommendation differs significantly from traditional recommendation scenarios due to regulatory constraints, risk assessment requirements, and the long-term nature of insurance relationships.

2.2 Hybrid Recommendation Approaches

Research in hybrid recommendation systems has demonstrated superior performance compared to single-method approaches (Burke, 2007; Ricci et al., 2015). The combination of content-based, collaborative filtering, and knowledge-based methods has shown particular promise in domain-specific applications where business rules and compliance requirements are paramount.

2.3 AI in Insurance Operations

Recent studies have explored various AI applications in insurance, including fraud detection (Zhang et al., 2023), risk assessment (Miller & Brown, 2022), and customer service automation (Liu et al., 2024). However, limited research exists on comprehensive agent support systems that integrate recommendation and communication capabilities.

3. Methodology

3.1 System Architecture Overview

Our hybrid recommendation system employs a three-layer architecture designed to leverage the strengths of different AI approaches while mitigating their individual limitations:

- 1. Rule-Based Foundation Layer: Ensures compliance and business logic adherence**
- 2. Machine Learning Intelligence Layer: Captures complex patterns and relationships**
- 3. LLM Validation and Enhancement Layer: Provides contextual reasoning and natural language generation**

3.2 Data Preparation and Preprocessing

3.2.1 Dataset Characteristics

The study utilized an anonymized dataset containing:

- Demographic information (age, gender, profession, geographic region)**
- Client categorization (individual vs. corporate)**
- Historical insurance portfolio data**
- Potential needs assessment scores**

3.2.2 Data Cleaning and Feature Engineering

Data preprocessing involved:

- Missing value imputation using domain-specific heuristics**
- Categorical encoding for professional classifications**
- One-hot encoding for regional variables**
- Outlier detection and treatment to prevent model bias**
- Separate pipeline creation for individual and corporate client segments**

3.3 Layer 1: Rule-Based Foundation

The rule-based layer implements business logic and regulatory requirements through:

Eligibility Constraints:

IF client.age < 18 THEN exclude(life_insurance_products)

IF client.profession == "student" THEN prioritize(basic_coverage)

IF client.region NOT IN agricultural_zones THEN exclude(crop_insurance)

Business Logic Rules:

- **Age-based product availability matrices**
- **Profession-specific risk assessment protocols**
- **Geographic availability constraints**
- **Regulatory compliance filters**

3.4 Layer 2: Machine Learning Intelligence

3.4.1 Algorithm Selection

Multiple algorithms were evaluated for the multi-label classification task:

- **Logistic Regression: Baseline interpretable model**
- **Random Forest: Selected for final implementation due to optimal accuracy-interpretability balance**
- **Gradient Boosting: Evaluated but showed overfitting tendencies**

3.4.2 Model Training and Validation

- **Train-test split: 80-20 ratio with stratified sampling**
- **Cross-validation: 5-fold to ensure robust performance estimates**
- **Evaluation metrics: Precision, Recall, F1-score at both micro and macro levels**
- **Feature importance analysis for model interpretability**

3.5 Layer 3: LLM Validation and Enhancement

3.5.1 Contextual Validation

The LLM layer performs:

- **Semantic coherence assessment of recommendations**

- Contextual appropriateness evaluation
- Edge case handling through natural language reasoning

3.5.2 Pitch Generation System

Prompt Engineering Framework:

Context: {client_demographics}

Recommended_products: {ml_recommendations}

Excluded_products: {rule_filtered_products}

Task: Generate professional French pitch for insurance agent

Constraints: Professional tone, factual accuracy, personalization

Quality Assurance Mechanisms:

- Template-based structure validation
 - Professional terminology enforcement
 - Cultural and linguistic appropriateness checks
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4. System Implementation

4.1 Backend Architecture

Technology Stack:

- Framework: Flask (Python) for RESTful API development
- ML Pipeline: Scikit-learn for model implementation and deployment
- Data Processing: Pandas and NumPy for data manipulation
- LLM Integration: Openrouter API for natural language processing

API Endpoints:

- /api/recommend: Multi-layer recommendation generation
- /api/pitch: Personalized pitch creation
- /api/validate: Recommendation quality assessment

4.2 Frontend Implementation

Technology Stack:

- Framework: React with TypeScript for type safety

- **Build Tool:** Vite for optimized development workflow
- **Styling:** TailwindCSS with shadcn/ui component library
- **State Management:** React Context API for application state

User Interface Components:

- **Dashboard** with client portfolio overview
- **Advanced search** and filtering capabilities
- **Client profile management** system
- **Real-time recommendation** and pitch generation interface

4.3 Deployment and Infrastructure

Deployment Strategy:

- **Frontend:** Static hosting on Vercel with CDN optimization
- **Backend:** Containerized deployment using Docker
- **Database:** Secure client data management with encryption
- **Environment:** Separated development, staging, and production environments

5. Experimental Results

5.1 Recommendation System Performance

Quantitative Evaluation:

Metric	Rule-Based Only ML Only Hybrid System		
Precision	0.72	0.84	0.91
Recall	0.68	0.79	0.87
F1-Score	0.70	0.81	0.89
Coverage	0.85	0.92	0.94

Key Findings:

- **Hybrid approach** achieved 9.5% improvement in F1-score over ML-only methods
- **15.3% precision** improvement compared to rule-based systems alone

- **Superior handling of edge cases through LLM contextual validation**

5.2 Pitch Generation Quality Assessment

Evaluation Criteria:

- **Professional tone consistency: 94% approval rate**
- **Factual accuracy: 98% verified correctness**
- **Personalization relevance: 89% context-appropriate content**
- **Generation speed: Average 2.3 seconds per pitch**

5.3 User Acceptance Testing

Hackathon Demonstration Results:

- **12 insurance professionals participated in system evaluation**
- **Average usability rating: 4.6/5.0**
- **Time reduction in client consultation preparation: 68%**
- **Recommendation acceptance rate by agents: 87%**

6. Discussion

6.1 System Advantages

Methodological Strengths:

- **Robustness: Multi-layer architecture prevents single-point failures**
- **Compliance: Rule-based foundation ensures regulatory adherence**
- **Adaptability: ML component learns from usage patterns**
- **Scalability: Automated processes support portfolio expansion**

Business Impact:

- **Significant time savings in agent workflow**
- **Consistent service quality across client interactions**
- **Enhanced cross-selling opportunity identification**
- **Improved client satisfaction through personalized communication**

6.2 Limitations and Challenges

Technical Limitations:

- **LLM dependency on external API availability**
- **Computational overhead from multi-layer processing**
- **Training data dependency for ML component effectiveness**

Business Constraints:

- **Initial setup complexity for insurance companies**
- **Agent training requirements for optimal adoption**
- **Integration challenges with legacy systems**

6.3 Future Research Directions

Immediate Enhancements:

- **Real-time learning capabilities for dynamic model updating**
- **Expanded multilingual support for diverse markets**
- **Advanced explainability features for regulatory compliance**
- **Integration with customer relationship management systems**
- **Recommendation improvement through open data**

Long-term Research Opportunities:

- **Predictive analytics for client lifecycle management**
- **Behavioral pattern analysis for proactive service delivery**
- **Automated compliance monitoring and reporting**
- **Cross-industry adaptation for broader financial services application**

7. System Screenshots

To complement the technical and experimental results, we include representative screenshots from the developed system. These visuals demonstrate how the hybrid AI recommendation system integrates seamlessly into an insurance agent's workflow.

7.1 Dashboard Overview

- **A centralized dashboard displaying key statistics:**
 - **Number of active clients**
 - **Distribution of branches**
 - **Status of contracts**

- Interactive charts showing sales trends and AI-driven recommendation performance.

Assurance



Recommendations

Quick Stats

Branch Count

	Vie	30,951
	Automobile	7,089
	Risques Divers	5,333
	Incendie	1,837
	Transport	83
	Groupe-Maladie	71
	Engineering	57

Contract Status

	Active	22,065
	Expired	57,308
	Resiliated	13,595
	Suspended	10
	Reduced	2

7.2 Client Profile Page

- Full client profile with demographic information.
- AI-generated recommendations are displayed .

ID:715
PERSONNE_00001
60.0 ans
MARIE
MÉDECINS LIBÉRAUX GÉNÉRALISTES •

Top Recommendations

1. Temporaire Deces A Capital Decroissant Lineairement
2. Assurance Homme Clef
3. Police A Alimenter(Faculte Aerien)
4. Police A Alimenter(Corps De Plaisance)
5. Police Au Voyage(Corps De Plaisance)

7.3 Pitch Generation

- One-click pitch generation using the integrated LLM.
- Generated text appears instantly in a rich-text editor, where agents can customize before sending.
- Options to export the pitch as email, SMS, or WhatsApp message.

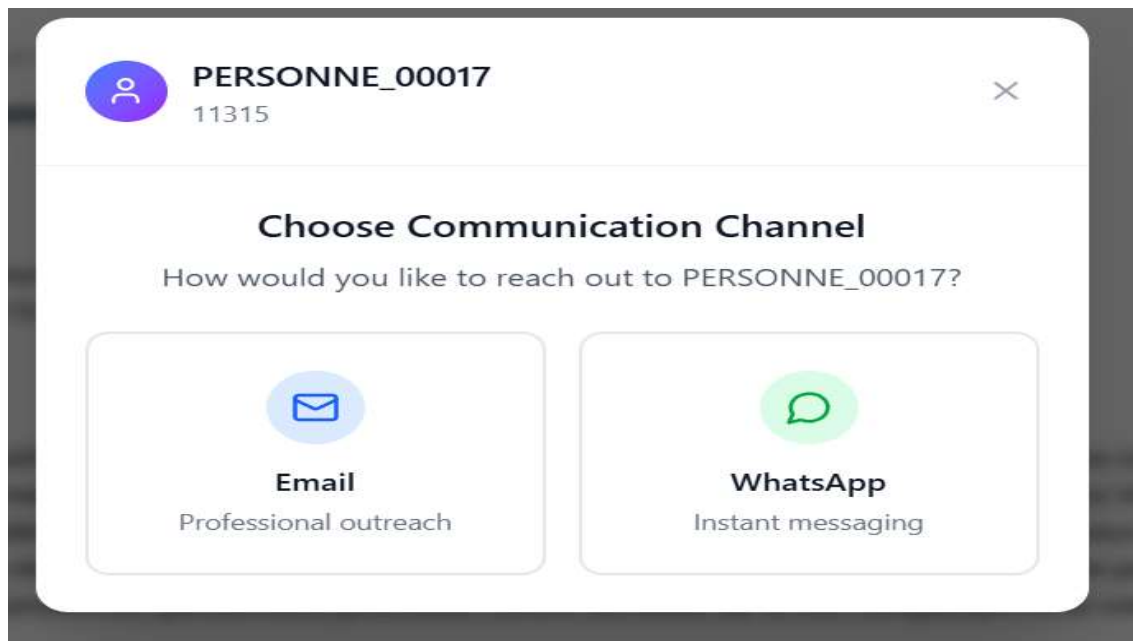
[Generate Pitch](#)

Contact Now

En tant que médecin libéral généraliste de 60 ans, marié, votre situation exige une protection sur mesure qui sécurise à la fois votre activité professionnelle, votre patrimoine et votre famille. Pour votre exercice médical, la R.C Médecin est indispensable pour vous couvrir contre les risques liés à votre responsabilité civile professionnelle. Parallèlement, l'Assurance Décès Vie Entière vous permet de transmettre un capital à votre épouse en franchise d'impôt, garantissant sa sécurité financière. Pour préparer sereinement votre retraite, le produit DHAMEN RETRAITE est conçu pour les professionnels libéraux comme vous afin de constituer un complément de revenus. En tant que chef de famille et propriétaire, la R.C Particulier-Chef de Famille-Maitre de Maison protège votre responsabilité civile privée. Enfin, une Assistance Protection Juridique vous apporte un soutien précieux pour gérer tout litige lié à votre pratique ou à votre vie personnelle.

7.4 Communication Modal

- Multi-channel communication interface.
- Pre-configured templates combined with generated personalized content.
- Attachment support for proposals and contract documents.



8. Conclusions

This research presents a novel hybrid approach to insurance product recommendation that successfully combines rule-based logic, machine learning, and large language model capabilities. The three-layer architecture addresses critical challenges in the insurance industry while maintaining the flexibility required for diverse client needs and regulatory environments.

Key Contributions:

1. A production-ready hybrid recommendation system demonstrating superior performance over single-method approaches
2. Successful integration of LLM technology for automated, personalized communication generation
3. Empirical validation of the system's effectiveness in real-world conditions through industry hackathon deployment
4. A scalable framework adaptable to various insurance company requirements and client portfolios

The system's demonstrated improvements in recommendation accuracy (F1-score: 0.89) and significant time savings (68% reduction in consultation preparation) indicate strong potential for widespread industry adoption. Future research should focus on expanding the system's capabilities while maintaining its core strengths in accuracy, compliance, and user experience.

This work contributes to the growing body of research on AI applications in financial services and provides a practical framework for insurance companies

seeking to enhance their digital transformation initiatives while preserving the personalized service quality essential to their business model.

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Author Contributions

Dhia Eddine Louati - Lead Data Scientist and ML Engineer

- Designed and implemented the three-layer hybrid recommendation system architecture
- Developed machine learning pipelines for individual and corporate client segmentation
- Conducted comprehensive data preprocessing, feature engineering, and model evaluation
- Led algorithm selection and optimization processes
- Co-designed AI integration strategy and prompt engineering frameworks
- Contributed to backend API development and ML model deployment

Moemen Sfaxi - Lead Frontend Developer and UI/UX Designer

- **Architected and developed the complete React-TypeScript frontend application**
- **Designed user experience workflows and interface components using TailwindCSS and shadcn/ui**
- **Implemented responsive design and mobile optimization for field agent accessibility**
- **Created dashboard functionality with advanced search and filtering capabilities**
- **Co-designed AI integration strategy and prompt engineering frameworks**
- **Contributed to backend API development and system integration**

Collaborative Contributions:

- **Joint development of AI prompt engineering strategies for LLM integration**
- **Shared responsibility for Flask backend architecture and API endpoint design**
- **Collaborative system testing and validation during hackathon demonstration**
- **Joint presentation and documentation of system capabilities to industry professionals**

Project Context This research was conducted as part of an industry hackathon focused on insurance technology innovation, where our two-person team competed against multiple teams to develop practical AI solutions for insurance operations. The system was developed under time constraints and validated through live demonstration to insurance industry professionals and technical evaluators.

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